1.1 Proof

Mathematics — OCR A-Level (H240) — Paper 1: Pure Mathematics

1. The Nature of Proof in Mathematics

Mathematical proof. A proof is a finite sequence of logically valid deductions that begins from accepted premises (axioms, definitions, and previously established theorems) and ends with the statement to be established. A proof leaves no possibility of the conclusion being false.

A single example — no matter how convincing — is *never* a proof. For instance, observing that $1^2 + 1 + 41 = 43$, $2^2 + 2 + 41 = 47$ and $3^2 + 3 + 41 = 53$ are all prime does not prove that $n^2 + n + 41$ is always prime (and in fact, it isn't — see §10). Empirical evidence in mathematics is a guide to a conjecture, not a substitute for deductive proof.

Axiom. A statement accepted as true without proof, taken as a starting point for a mathematical system (e.g. the associative law of addition, or the axiom that through any two distinct points there passes exactly one straight line).

Theorem. A statement that has been proved from the axioms and previously established results. A *lemma* is a smaller supporting theorem used to prove a larger one; a *corollary* is an immediate consequence of a theorem.

Logical and Set Notation

OCR H240 expects fluency with the following symbols:

Symbol	Read as	Meaning

$\forall$	for all	Universal quantifier — the statement holds for every element.
$\exists$	there exists	Existential quantifier — at least one element satisfies the statement.
$\Rightarrow$	implies	If the left-hand side is true, the right-hand side is true.
$\Leftrightarrow$	if and only if	Each side implies the other (logical equivalence).
$\neg$	not	Negation — the logical opposite.
$\in$	is an element of	Membership in a set.
$\mathbb{N}$	the naturals	$\{0, 1, 2, 3, \ldots\}$ (or $\{1, 2, 3, \ldots\}$ depending on convention).
$\mathbb{Z}$	the integers	$\{\ldots, -2, -1, 0, 1, 2, \ldots\}$.
$\mathbb{Q}$	the rationals	Numbers expressible as $\frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b \neq 0$.
$\mathbb{R}$	the reals	All rationals together with all irrationals.
$\therefore$	therefore	Introduces a conclusion.
$\square$ / QED	end of proof	"Quod erat demonstrandum" — "which was to be demonstrated".

OCR examiner note. Marks for proof are awarded for the *structure* of the argument as well as the algebra. Clearly stated quantifiers (e.g. "for all $n \in \mathbb{Z}$") and explicit concluding sentences ("hence the statement is true") are routinely the difference between full marks and a single mark lost for "insufficient justification".

2. Proof by Deduction (Direct Proof)

Proof by deduction. Starting from accepted premises (definitions, axioms, and already-proved theorems), a chain of logical implications leads to the required conclusion. Each step must be a valid inference from the previous one.

Useful Algebraic Representations

- Even number: $2n$ where $n \in \mathbb{Z}$.
- Odd number: $2n+1$ where $n \in \mathbb{Z}$.
- Consecutive integers: $n, n+1, n+2, \ldots$.
- Consecutive odd numbers: $2n+1, 2n+3, 2n+5, \ldots$.
- Multiple of 3: $3n$ where $n \in \mathbb{Z}$.
- Rational number: $\dfrac{p}{q}$ where $p \in \mathbb{Z}$, $q \in \mathbb{Z} \setminus \{0\}$.

Worked Example 1 — Sum of two consecutive odd numbers

Claim. For every integer n, the sum of two consecutive odd numbers is divisible by 4.

```
Let the two consecutive odd numbers be  2n+1  and  2n+3,      // n ∈ ℤ
Sum  =  (2n+1) + (2n+3)
      =  4n + 4                                                // collect like
      =  4(n+1).                                               // factor out 4
Since n ∈ ℤ,  n+1 ∈ ℤ, so 4(n+1) is an integer multiple of 4.
∴ the sum is divisible by 4.                                   // □
```

Worked Example 2 — $n^2 + n$ is always even

Claim. For every $n \in \mathbb{Z}$, $n^2 + n$ is even.

Proof. $n^2 + n = n(n+1)$. This is the product of two consecutive integers, exactly one of which must be even, so the product is even. $\square$

Worked Example 3 — Product of two odd numbers is odd

Let the odd numbers be $2a+1$ and $2b+1$ with $a, b \in \mathbb{Z}$.

$$(2a+1)(2b+1) = 4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1.$$

Since $2ab + a + b \in \mathbb{Z}$, this is of the form $2k + 1$ and hence odd. $\square$

3. Proof by Exhaustion

Proof by exhaustion. Divide the problem into a *finite* list of cases that together cover every possibility, then verify the statement holds in each case separately.

Exhaustion only works when the number of cases is finite (or finite after a sensible reduction using modular arithmetic). It is powerful when combined with the observation that every integer n leaves one of a finite set of remainders on division by some fixed modulus.

Worked Example 4 — $n^2 + 2$ is never divisible by 4

Claim. For every $n \in \mathbb{Z}$, $n^2 + 2$ is not divisible by 4.

Proof. Every integer satisfies exactly one of $n \equiv 0, 1, 2, 3 \pmod 4$. We check each case.

```
// Case analysis on n mod 4
1. If n ≡ 0 (mod 4):  n = 4k          ⇒  n² + 2 = 16k² + 2        = 4(4k²) + 2
2. If n ≡ 1 (mod 4):  n = 4k+1        ⇒  n² + 2 = 16k² + 8k + 3  = 4(4k²+2k) +
3. If n ≡ 2 (mod 4):  n = 4k+2        ⇒  n² + 2 = 16k² + 16k + 6 = 4(4k²+4k+1)
4. If n ≡ 3 (mod 4):  n = 4k+3        ⇒  n² + 2 = 16k² + 24k + 11= 4(4k²+6k+2)
5. Every case leaves remainder 2 or 3 when divided by 4, never 0.
6. ∴ n² + 2 is never divisible by 4.                                        //
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Worked Example 5 — Every integer cube has last digit in a restricted set

Claim. For $n \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, n^3 ends in one of $0, 1, 8, 7, 4, 5, 6, 3, 2, 9$.

n	0	1	2	3	4	5	6	7	8	9
n^3	0	1	8	27	64	125	216	343	512	729
last digit	0	1	8	7	4	5	6	3	2	9

All ten cases checked. □

OCR exam tip. State the cases *before* you start checking them. Examiners reward a clear structural sentence such as "Every integer satisfies exactly one of $n \equiv 0, 1, 2, 3 \pmod 4$; we consider each case." If you simply compute without declaring the case split, you can lose a structural mark even when the algebra is correct.

4. Proof by Counter-Example

Counter-example. A single specific instance in which a universal claim fails. One counter-example is sufficient to disprove a statement of the form "for every x, $P(x)$ is true".

Worked Example 6 — Disproving "$2^n - 1$ is prime for all $n \geq 2$"

Claim (false). For every integer $n \geq 2$, $2^n - 1$ is prime.

Disproof. Take $n = 4$. Then $2^4 - 1 = 15 = 3 \times 5$, which is composite. Hence the claim fails. □

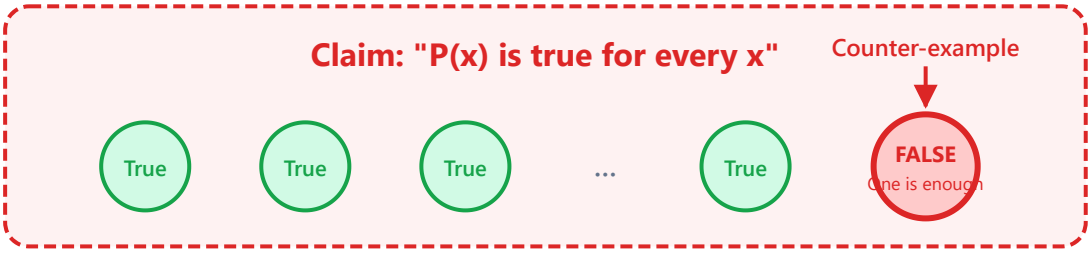
(Remark: $2^n - 1$ is called a Mersenne number; it is known that $2^n - 1$ can only be prime if n is itself prime, but even primality of n is not sufficient — e.g. $2^{11} - 1 = 2047 = 23 \times 89$.)

Worked Example 7 — Disproving "sum of two irrationals is irrational"

Take $a = \sqrt{2}$ and $b = -\sqrt{2}$. Both are irrational, but $a + b = 0 \in \mathbb{Q}$. $\square$

Worked Example 8 — Disproving "$n^2 > n$ for every $n \in \mathbb{Z}$"

Take $n = 0$. Then $n^2 = 0 \not> 0$. $\square$



OCR exam tip. When a question says "disprove", "show that the statement is false" or "show by counter-example", a single specific numerical counter-example with a short justification gets the marks. Don't waste time on a general argument — pick the simplest value that works. Try $n = 0, 1, -1, 2$ first.

5. Proof by Contradiction

Proof by contradiction (reductio ad absurdum). To prove a statement S, assume $\neg S$ is true. Show that $\neg S$, combined with known facts, leads to a contradiction (a statement that is self-contradictory or contradicts an established result). Since a contradiction cannot hold, the assumption $\neg S$ must be false, so S is true.

Structure of a Proof by Contradiction

1. **Assumption.** Write "Assume for contradiction that ..." and state the negation of the claim.

2. **Deduction.** Apply valid logical steps from this assumption.
3. **Contradiction.** Identify the impossibility explicitly — e.g. "but this contradicts our assumption that $\frac{a}{b}$ was in lowest terms".
4. **Conclusion.** "Hence the assumption was false, and the original statement is true." Write □.

Landmark Example — $\sqrt{2}$ is irrational

Claim. There is no rational number whose square is 2.

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// Proof by contradiction that √2 ∉ ℚ.
 1. Assume, for contradiction, that √2 ∈ ℚ.
 2. Then √2 = a/b with a, b ∈ ℤ, b ≠ 0,
      and a/b is written in lowest terms (gcd(a,b) = 1).          // WLOG we can
 3. Squaring:           2   =   a²/b²
 4.                     a²  =   2 b².                                  // (*)
 5. So a² is even  ⇒   a is even        // since the square of an odd number is
 6. Write a = 2k,  k ∈ ℤ.
 7. Substitute into (*):   (2k)² = 2 b²
 8.                         4 k² = 2 b²
 9.                         b²   = 2 k².                            // so b² is
10. So b² is even  ⇒   b is even.
11. But then 2 divides both a and b, so gcd(a,b) ≥ 2.
12. This CONTRADICTS step 2, where gcd(a,b) = 1.
13. ∴ the assumption in step 1 is false, so √2 is irrational.       // □
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Why step 5 is valid: if a were odd, say $a = 2m + 1$, then $a^2 = 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1$, which is odd. So a^2 even forces a even. (This is itself a small proof by contrapositive — see §6.)

Landmark Example — Euclid's proof that there are infinitely many primes

Claim. There are infinitely many prime numbers.

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// Euclid, c. 300 BC. Proof by contradiction.
 1. Assume, for contradiction, that there are only finitely many primes.
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2. List them all:   p₁, p₂, …, p_n.
3. Consider N = p₁ · p₂ · … · p_n + 1.
4. N > 1, so N has at least one prime factor — call it q.
5. q is on the list, so q = p_i for some i ≤ n.
6. Then q divides (p₁ · … · p_n),                    // q is one of the factors
     and q divides N (by choice of q).
7. ⇒  q divides  N − p₁·…·p_n  =  1.                // contradiction
8. But no prime divides 1.  CONTRADICTION.
9. ∴ the list of primes cannot be finite: there are infinitely many.  // □
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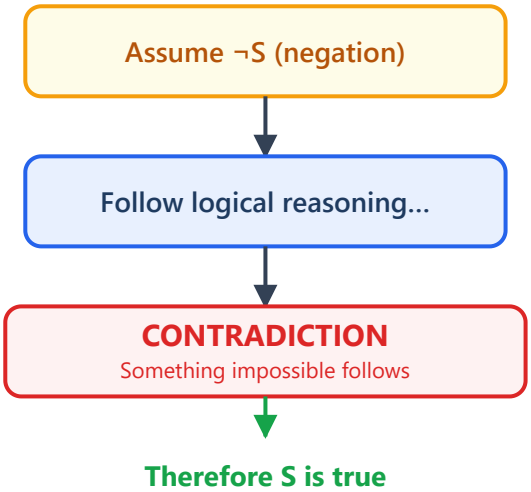

OCR examiner note — proof by contradiction. Markers look for four explicit features: (i) the word "assume" (or "suppose") introducing the negation; (ii) the correct negation of the claim; (iii) a visible contradiction (usually underlined or signposted with "but this contradicts…"); (iv) the closing sentence "Hence the assumption is false, and S is true". Omitting the closing sentence routinely costs one mark.

6. Contrapositive Proof

Contrapositive. The contrapositive of the implication $P \Rightarrow Q$ is $\neg Q \Rightarrow \neg P$. A statement and its contrapositive are logically equivalent: proving one proves the other.

Contrapositive proof is useful when $\neg Q$ gives a more tractable algebraic form than P does. It is technically a direct proof — of $\neg Q \Rightarrow \neg P$ — but is often listed alongside contradiction because it also proceeds

"from the opposite end".

Worked Example 9 — "If n^2 is even then n is even"

Let P: "n^2 is even" and Q: "n is even". The contrapositive is "$\neg Q \Rightarrow \neg P$", i.e. "if n is odd then n^2 is odd".

Proof of contrapositive. Suppose n is odd, so $n = 2k + 1$ with $k \in \mathbb{Z}$. Then

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is of the form $2m + 1$ and hence odd. So $\neg Q \Rightarrow \neg P$. By the equivalence of an implication with its contrapositive, $P \Rightarrow Q$. $\square$

7. Logic Essentials — Converse, Inverse, Contrapositive

Starting from an implication $P \Rightarrow Q$, there are three related statements students routinely confuse:

Name	Form	Equivalent to original?
Original	$P \Rightarrow Q$	—
Converse	$Q \Rightarrow P$	**Not** (in general)
Inverse	$\neg P \Rightarrow \neg Q$	**Not** (in general)
Contrapositive	$\neg Q \Rightarrow \neg P$	**Yes** — always equivalent

Converse. The converse of $P \Rightarrow Q$ is $Q \Rightarrow P$. The truth of the converse is an independent question.

Implication vs equivalence. $P \Rightarrow Q$ says only "if P then Q". $P \Leftrightarrow Q$ says both $P \Rightarrow Q$ and $Q \Rightarrow P$ — i.e. both the original and its converse hold.

Truth Tables

A truth table lists every combination of truth values for the atomic statements and evaluates the compound.

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$ (converse)	$\neg Q \Rightarrow \neg P$ (contrapositive)
T	T	T	T	T
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

Observe that the column for $P \Rightarrow Q$ matches the column for its contrapositive in every row (proving equivalence), but differs from the converse in rows 2 and 3.

Worked Example 10 — Distinguishing converse and implication

Statement S: "If $x = 2$ then $x^2 = 4$". This is true for every x.

Converse: "If $x^2 = 4$ then $x = 2$". This is *false* — take $x = -2$, then $x^2 = 4$ but $x \neq 2$.

So S is true but its converse is not: the equivalence $x = 2 \Leftrightarrow x^2 = 4$ fails.

Common mistake. When asked to prove "if and only if" ($\Leftrightarrow$), you must prove *both directions*. Write "($\Rightarrow$)" and then "($\Leftarrow$)" as two separate sub-proofs. Proving only one direction scores at most half the marks.

8. Disproving by Contradiction

A statement S can be disproved by contradiction: assume S is true and derive a contradiction; this forces S to be false.

Worked Example 11 — Disprove "there is a largest natural number"

Proof. Assume, for contradiction, that there is a largest natural number $M \in \mathbb{N}$. Then $M + 1$ is also a natural number, and $M + 1 > M$. This contradicts M being the largest. Hence no largest natural number exists. $\square$

Worked Example 12 — Disprove "every prime is odd"

If every prime were odd, then 2 — which is prime — would be odd. But 2 is even. Contradiction. $\square$

(This is really a counter-example, showing that counter-example and disproof-by-contradiction are two sides of the same coin when the universally quantified claim has a computable failing instance.)

9. Proof by Induction (scope note)

Proof by induction establishes a statement $P(n)$ for all $n \in \mathbb{Z}^+$ (or $n \geq n_0$) in two steps: verify $P(1)$ (the *base case*), and show that $P(k) \Rightarrow P(k+1)$ for every $k \geq 1$ (the *inductive step*).

Scope for OCR H240. Induction is *not* part of the OCR A-Level Mathematics H240 specification. It appears in OCR A-Level Further Mathematics (H245) under the Pure core. You are not expected to prove results by induction in any H240 paper. If you are continuing to Further Maths, see the H245 notes; otherwise you may safely focus on the four techniques in §2–§8.

10. Worked Exam-Style Questions (H240 style)

Question A — Product of two rationals is rational (2 marks)

Prove that the product of two rational numbers is rational.

Solution. Let $x, y \in \mathbb{Q}$. By definition of rational number, write

$x = \frac{a}{b}$ and $y = \frac{c}{d}$, with $a, c \in \mathbb{Z}$ and $b, d \in \mathbb{Z} \setminus \{0\}$.

Then

$$xy = \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

Since $a, c \in \mathbb{Z}$ we have $ac \in \mathbb{Z}$; since b, d are non-zero integers we have $bd \in \mathbb{Z} \setminus \{0\}$. Therefore xy is of the form $\frac{\text{integer}}{\text{non-zero integer}}$, i.e. $xy \in \mathbb{Q}$. $\square$

Marking notes.

- M1 for writing both rationals in $\frac{p}{q}$ form with the integer conditions stated explicitly.
- A1 for identifying that $ac \in \mathbb{Z}$ and $bd \neq 0$, and concluding the product is rational.

Question B — Rational + irrational is irrational (4 marks)

Prove by contradiction that the sum of a rational number and an irrational number is irrational.

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// Set-up
 1. Let r ∈ ℚ  and  i ∈ ℝ \ ℚ (irrational).  Consider s = r + i.
 2. Assume, for contradiction, that s ∈ ℚ.

// Deduction
 3. Write r = a/b  with a ∈ ℤ, b ∈ ℤ \ {0},
     and s = p/q with p ∈ ℤ, q ∈ ℤ \ {0}.
 4. Then i = s − r = p/q − a/b
                   = (pb − aq) / (qb).
 5. Numerator pb − aq ∈ ℤ; denominator qb ∈ ℤ \ {0}.
 6. ∴ i ∈ ℚ.                                          // (*)

// Contradiction
 7. But i was assumed irrational, so i ∉ ℚ.  Contradicts (*).
 8. ∴ the assumption in step 2 is false: s ∉ ℚ, i.e. s is irrational.   // □
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Marking notes.

- B1 for the opening contradiction assumption ("assume $r + i$ is rational").
- M1 for expressing r and $r + i$ as fractions of integers with non-zero denominators.

- M1 for rearranging to show i would be rational.
- A1 for identifying the contradiction and stating the conclusion.

Question C — Disprove "$n^2 + n + 41$ is always prime" (2 marks)

Disprove the statement: for every $n \in \mathbb{N}$, $n^2 + n + 41$ is prime.

Solution. Take $n = 41$. Then

$$n^2 + n + 41 = 41^2 + 41 + 41 = 41(41 + 1 + 1) = 41 \times 43,$$

which is composite (it has factors 41 and 43). The statement is therefore false. $\square$

(Remark: this famous polynomial, due to Euler, does produce primes for $n = 0, 1, 2, \ldots, 39$, but fails at $n = 40$ — where $40^2 + 40 + 41 = 1681 = 41^2$ — and at $n = 41$ as above. It is a textbook example of why empirical evidence is not proof.)

Marking notes.
- M1 for a specific value of n that produces a composite.
- A1 for showing the factorisation and concluding that the statement is false.

Question D — Difference of two consecutive squares is odd (3 marks)

Prove by direct deduction that the difference between the squares of two consecutive integers is always odd.

Solution. Let the two consecutive integers be n and $n + 1$, where $n \in \mathbb{Z}$. Their squares are n^2 and $(n + 1)^2$. Consider the larger minus the smaller:

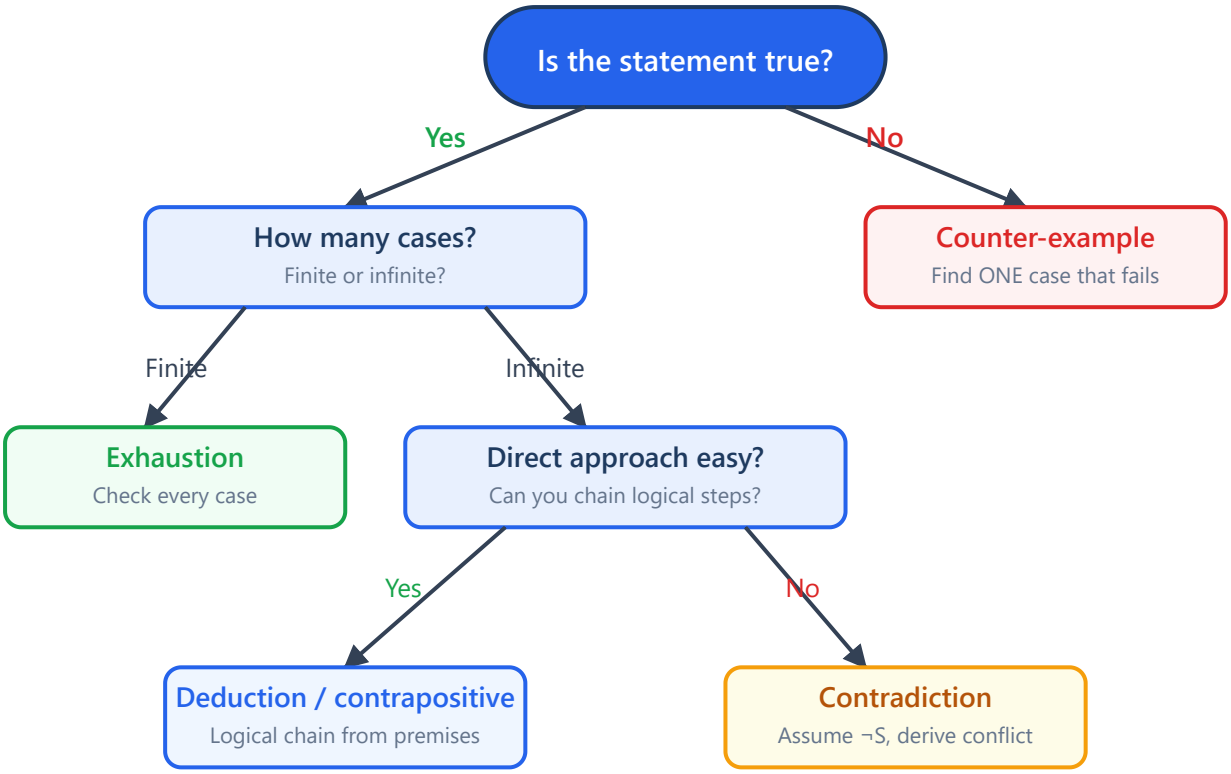
$$(n + 1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1.$$

Since $n \in \mathbb{Z}$, $2n + 1$ is of the form "even + 1" and is therefore odd. $\square$

Marking notes.
- M1 for algebraic set-up using n and $n + 1$ and forming the difference.
- M1 for expanding to $2n + 1$.
- A1 for the closing justification that $2n + 1$ is odd for any integer n.

11. Choosing the Right Technique

If the claim says...	Start with...	Key signal
"Prove for all n..."	Deduction	General algebraic form ($2n, 2n+1$, etc.)
"Prove for $n = 1, 2, \ldots, 10$..."	Exhaustion	Finite list — just check them all
"Prove n^2 has a property for every n..."	Deduction or exhaustion via $n \bmod k$	Reduce infinitely many integers to finitely many residue cases
"Disprove / show false..."	Counter-example	One value is enough
"Prove $\sqrt{\text{something}}$ is irrational"	Contradiction	Irrationality is hard to show directly
"Prove there are infinitely many..."	Contradiction (assume finitely many)	Negating "infinite" gives you something concrete to work with
"Prove $P \Rightarrow Q$ where P is awkward"	Contrapositive ($\neg Q \Rightarrow \neg P$)	If $\neg Q$ has a nicer algebraic form
"Prove $P \Leftrightarrow Q$"	Both directions separately	Two proofs: $P \Rightarrow Q$ and $Q \Rightarrow P$



12. OCR Exam Strategy Summary

Key exam strategies for proof questions (H240).

- **Read the verb carefully.** "Prove", "show", "demonstrate" call for a full argument; "disprove", "show false" want a counter-example.
- **Define your variables.** Examiners routinely deduct a mark for a missing "where $n \in \mathbb{Z}$" or similar.
- **For deduction** — introduce the algebraic form ($2n+1$, $3k$, $\frac{a}{b}$), manipulate, and conclude with the required form made explicit.
- **For exhaustion** — list *all* cases up front, then check each. If you use "$n \equiv 0, 1, 2, 3 \pmod 4$", say so.
- **For counter-example** — a single specific value, followed by the computation and a one-line conclusion "the statement is false".

- **For contradiction** — always open "assume for contradiction that ...". Finish with the explicit identification of the contradiction and the sentence "hence the original statement is true".
- **For contrapositive** — say so: "we prove the contrapositive, which is equivalent". Examiners want to see you know why this is legitimate.
- **End every proof** with a concluding sentence that directly addresses the original claim, and close with □ or QED. This routinely gains a "completeness" mark.

Common pitfalls flagged in OCR mark schemes.

- Using the same letter (e.g. n) for two different quantities in the same proof.
- Writing "a is even, therefore a^2 is even" when you needed the *reverse* ("a^2 even ⇒ a even"). These statements look similar but are different implications; the reverse requires the contrapositive argument.
- Forgetting the condition $b \neq 0$ when writing a rational as $\frac{a}{b}$.
- Writing "$\sqrt{2} = \frac{a}{b}$" without declaring "in lowest terms" — the proof fails without that assumption.
- Confusing a statement with its converse (see §7).